

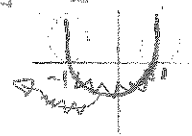
Technique Mastering Exercises

①

A - Absolute Value

1. a) $|x-2| = \begin{cases} x-2 & \text{if } x-2 \geq 0 \\ -(x-2) & \text{if } x-2 < 0 \end{cases}$
 $= \begin{cases} x-2 & \text{if } x \geq 2 \\ -x+2 & \text{if } x < 2 \end{cases}$

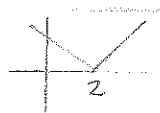
b) $|3+6x| = \begin{cases} 3+6x & \text{if } 3+6x \geq 0 \\ -(3+6x) & \text{if } 3+6x < 0 \end{cases}$
 $= \begin{cases} 3+6x & \text{if } x \geq -\frac{1}{2} \\ -3-6x & \text{if } x < -\frac{1}{2} \end{cases}$

c) $|x^2-1| = \begin{cases} x^2-1 & \text{if } x^2-1 \geq 0 \\ -(x^2-1) & \text{if } x^2-1 < 0 \end{cases}$
 $= \begin{cases} x^2-1 & \text{if } x \geq 1 \text{ or } x \leq -1 \\ -x^2+1 & \text{if } -1 < x < 1 \end{cases}$
 $x^2-1 = (x-1)(x+1)$


d) $|x^2+4| = \begin{cases} x^2+4 & \text{if } x^2+4 \geq 0 \\ -(x^2+4) & \text{if } x^2+4 < 0 \end{cases}$
 $= x^2+4 \text{ for all } x \in \mathbb{R}$


2 (a) $|x-2| = 0$
 $\Rightarrow x-2 = 0 \Rightarrow x = 2$

(b) $|x-2| > 0$
 $\Rightarrow x-2 > 0 \text{ or } x-2 < 0$
 $\Rightarrow x > 2 \text{ or } x < 2 \Rightarrow x \neq 2$
 $\Rightarrow x \in \mathbb{R} \setminus \{2\}$


(c) $|x-2| < 0$
 $\Rightarrow x \in \emptyset \text{ (no solution)}$


(d) $|x-2| < 1$
 $\Rightarrow -1 < x-2 < 1 \Rightarrow -1+2 < x-2+2 < 1+2$
 $\Rightarrow 1 < x < 3 \Rightarrow x \in (1, 3)$

(e) $|2x| < 7$
 $\Rightarrow -7 < 2x < 7 \Rightarrow -\frac{7}{2} < x < \frac{7}{2}$
 $\Rightarrow x \in (-\frac{7}{2}, \frac{7}{2})$

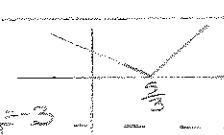
(f) $|2x-4| > 8$
 $\Rightarrow 2x-4 > 8 \text{ or } 2x-4 < -8$
 $\Rightarrow 2x > 12 \text{ or } 2x < -4$
 $\Rightarrow x > 6 \text{ or } x < -2$
 $\Rightarrow x \in (-\infty, -2) \cup (6, \infty)$

(g) $|x-3| \leq 6$
 $\Rightarrow -6 \leq x-3 \leq 6 \Rightarrow -6+3 \leq x \leq 6+3$
 $\Rightarrow -3 \leq x \leq 9 \Rightarrow x \in [-3, 9]$

(h) $|2x+5| > 6$
 $\Rightarrow 2x+5 > 6 \text{ or } 2x+5 \leq -6$
 $\Rightarrow 2x > 1 \text{ or } 2x \leq -11$
 $\Rightarrow x > \frac{1}{2} \text{ or } x \leq -\frac{11}{2}$
 $\Rightarrow x \in (-\infty, -\frac{11}{2}] \cup [\frac{1}{2}, \infty)$

(i) $|1-2x| = 3$ See p A7 Stewart
 $\Rightarrow 1-2x = 3 \text{ or } 1-2x = -3$
 $\Rightarrow -2x = 2 \text{ or } -2x = -4$
 $\Rightarrow x = -1 \text{ or } x = 2$
 $\therefore x \in \{-1, 2\}$

(j) $|3-7x| \leq 6$
 $\Rightarrow -6 \leq 3-7x \leq 6 \Rightarrow -9 \leq -7x \leq 3$
 $\Rightarrow -\frac{9}{-7} \geq x \geq -\frac{3}{-7} \Rightarrow -\frac{3}{7} \leq x \leq \frac{9}{7}$
 $\Rightarrow x \in [-\frac{3}{7}, \frac{9}{7}]$

(k) $|5-3x| = -3$
 $\therefore x \in \emptyset \text{ (no solution)}$


(l) $|1-5x-8| > 12$
 $\Rightarrow -5x-8 > 12 \text{ or } -5x-8 \leq -12$
 $\Rightarrow -5x > 20 \text{ or } -5x \leq -4$
 $\Rightarrow x < -\frac{20}{5} \text{ or } x \geq \frac{4}{5}$
 $\Rightarrow x < -4 \text{ or } x \geq \frac{4}{5}$
 $\Rightarrow x \in (-\infty, -4) \cup (\frac{4}{5}, \infty)$

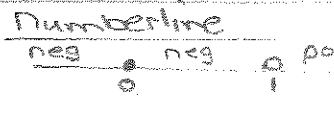
(m) $|x| = |-3| = 3$
 $\Rightarrow x = 3 \text{ or } x = -3$
 $\Rightarrow x \in \{-3, 3\}$

(n) $|3x| = |2x+1|$
 $\Rightarrow (3x)^2 = (2x+1)^2$
 $\Rightarrow 9x^2 = 4x^2 + 4x + 1$
 $\Rightarrow 5x^2 - 4x - 1 = 0$
 $\Rightarrow (5x+1)(x-1) = 0$
 $\Rightarrow x = -\frac{1}{5} \text{ or } x = 1$
 $\Rightarrow x \in \{-\frac{1}{5}, 1\}$

• p A7 Stewart
 $\sqrt{a^2} = |a|$
 and
 • Studyguide
 Unit 1.2
 Theorem
 $|x| \leq |y|$
 $\Rightarrow x^2 \leq y^2$

(o) $\frac{x-1}{|2x|} \leq 0$ Numberline
 neg pos

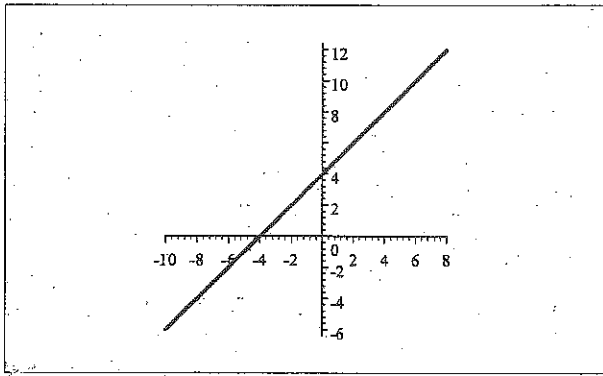
 $\Rightarrow x \in (-\infty, 0) \cup (0, 1]$

(p) $\frac{|2x|}{x-1} \leq 0$ Numberline
 neg pos

 $\Rightarrow x \in (-\infty, 1)$

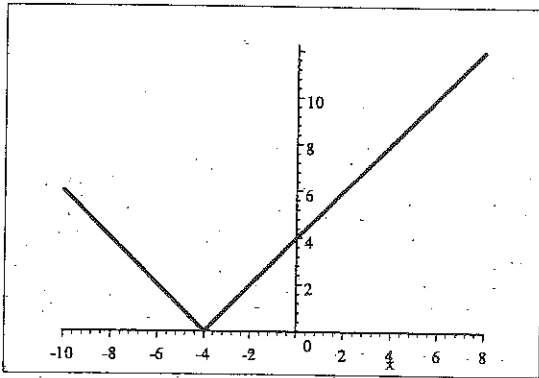
A - Absolute Value

②

3(a)

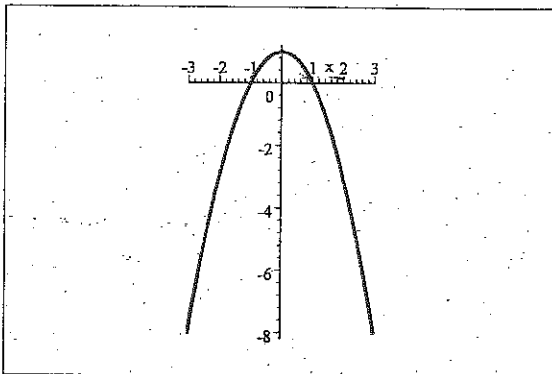


$$y = x + 4$$

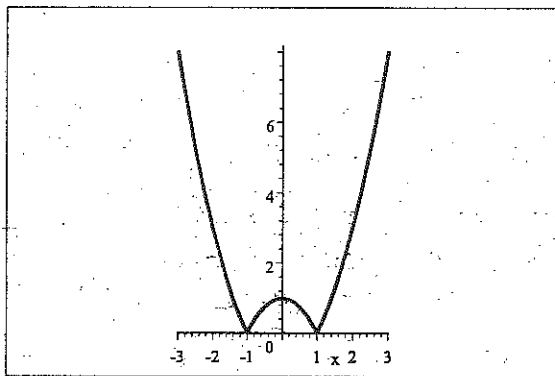


$$y = |x + 4|$$

3(b)

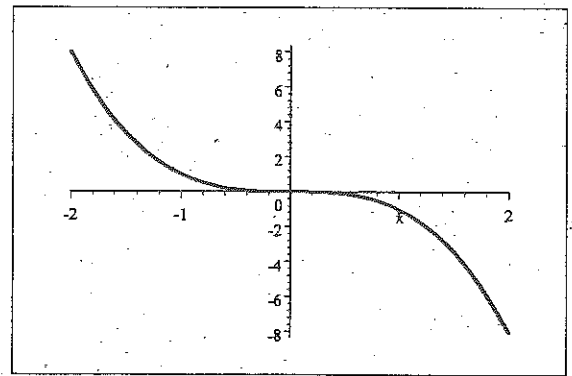


$$y = 1 - x^2$$

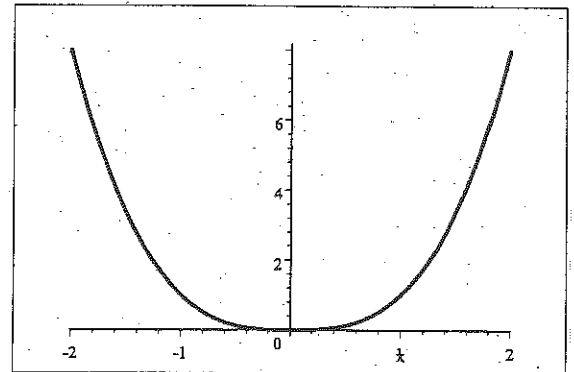


$$y = |1 - x^2|$$

3(c)

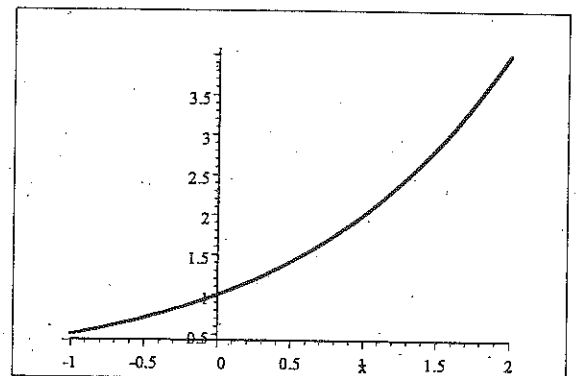


$$y = -x^3$$

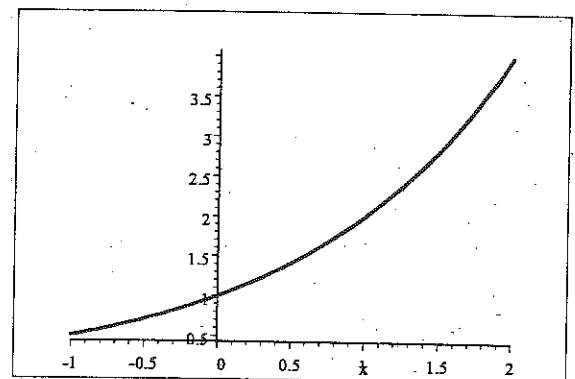


$$y = |-x^3|$$

3(d)



$$y = 2^x$$



$$y = |2^x|$$